

IMPROVED EFFICIENCY OF FOG AND CLOUD COMPUTING PARADIGMS

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Abstract—For effective energy management in smart grids, this research suggests a FOG computing architecture that bridges the gap between the edge and CLOUD. Facing data congestion in both traditional edge and CLOUD computing, the system utilizes a two-tiered approach. By preprocessing and filtering sensor data from buildings, edge nodes reduce latency and network traffic. The FOG layer in turn effectively filters data for potential CLOUD processing while doing real time analysis and decision-making depending on immediate needs. With latency reduction of up to 98.7%, the simulation demonstrates significant benefits over CLOUD only approaches and enables real time energy adjustments that optimize resource utilization. Overall, scalability and efficient data handling overcome CLOUD limitations, while enhanced security minimizes reliance on a central CLOUD. This distributed brain for the grid paves the way for a smarter, more responsive energy landscape, powered by intelligence at the edge.

Keywords—FOG computing, CLOUD computing, smart grids, energy management, iFogSim, latency reduction, real-time analysis

I. INTRODUCTION

FOG computing is used as a middle layer between the edge and the CLOUD to handle data processing without the delays of sending data to the CLOUD and without the limitations of edge computing. It is beneficial for industries like power grids that generate a large amount of data [5] from multiple sources. The FOG server has a program that decides whether to send data to the CLOUD or not, acting as a filter for smaller scale data [3] that does not need to be constantly sent to the CLOUD. The goal is to keep the overall system execution time low.

- Avoid data congestion in the FOG server.
- Fix the power consumption problem faced by FOG servers.
- Find a better solution to encrypting and decrypting the data without compromising its security to make processing easier for the FOG server system.

These are the most prevalent issues in FOG computing that would be addressed in this research. [2]

This study explores the effectiveness of FOG and CLOUD computing in managing energy resources in smart grids.

Traditional computing methods struggle to handle substantial amounts of data [1,11] from smart grid sensors. A new FOG computing architecture is proposed, combining edge and CLOUD computing to improve energy management. The architecture involves edge nodes preprocessing data to reduce congestion and execution time, with the FOG layer making real-time decisions and filtering data for potential CLOUD processing. Through simulations, it is evident that the architecture significantly reduces execution time and improves scalability and security compared to CLOUD-only methods. This research aims to advance intelligent energy management systems for a more efficient and responsive energy landscape.

Motivation

The increasing demand for energy management in smart grids [7] within reasonable limits can be met by applying novel solutions that focus on the constraints arising from CLOUD computing-based methods. While there are other options for processing renewable energy data, including CLOUD power onboard EVs – a trend likely to grow in the coming years through grid-responsive charge control to match it against flare spots and excess generation available on grids [8]. As one can understand, the network congestion caused by massive flow of data will impede real time decision making, which is central to energy optimization.

Research Contributions

The research makes significant contributions to the understanding and advancement of FOG and CLOUD computing paradigms, particularly in the context of smart grid energy management. The following key contributions are highlighted:

Introduction of FOG Computing Architecture:

The paper introduces a unique architecture for FOG computing that improves communication between edge devices and the CLOUD. This two-tier approach combines the FOG layer with edge nodes to address energy management challenges in smart grids and reduce data congestion and delays during peak hours.

Mitigation of Common FOG Computing Challenges:

The study highlights typical issues in the FOG network such as data congestion, power usage, and data security. Proposed solutions aim to address these challenges by

preventing server overload, reducing power consumption, and improving data encryption for enhanced security.

Empirical Analysis and Comparison of FOG and CLOUD Computing:

This paper performs an empirical analysis and comparison of performance characteristics for FOG computing and CLOUD computing architectures in terms of execution time and network usage through simulation experiments. Based on the results from research, it can be inferred that FOG computing has a lot of advantage over CLOUD computing, especially in scenarios featuring the increasing density of devices at the edge of the network.

Use of iFogSim in Simulation and Modelling:

The study uses iFogSim, a software framework made for simulating FOG computing, to analyse the performance of FOG computing architecture and applications in various scenarios. By studying how different parameters impact behaviour and resource scaling, the goal is to improve energy efficiency and performance compared to other systems.

Sustainability aspect:

The proposed FOG computing architecture for smart grid management enhances real-time decision-making and data handling while contributing to a more sustainable energy landscape. By reducing reliance on CLOUD computing for data processing tasks and distributing intelligence to the FOG layer, it significantly lowers energy consumption associated with data transmission and CLOUD processing. This shift reduces the overall power usage of the grid, resulting in more energy-efficient computing. Furthermore, real-time energy adjustments enabled by the system optimize resource utilization and minimize energy waste. This approach aligns with the goals of green technologies, promoting a more sustainable, energy-efficient, and responsive smart grid.

II. RELATED STUDIES

1. A thorough analysis conducted between 2015-2020 [12] reviewed journals, conferences, and dissertations on FOG computing, systematically identifying and cataloguing existing simulation tools. The primary aim was to map the current landscape of these tools and shed light on the challenges inherent in their development. This valuable exploration not only reveals the available resources but also illuminates the intricacies involved in building software for FOG simulation.
2. Thorough systematic literature reviews [13,14] explore resource management strategies for FOG computing in response to the urgent need for a centralized evaluation. The review addresses the difficulties presented by limited resources, diversity, and the ever-changing nature of FOG environments. By organizing resource management techniques into six main categories, including program management, scheduling, task execution, allocation, provisioning, and load balancing, a structured framework is provided for comparing different approaches. This in-depth analysis considers essential factors such as performance measurements, methodologies, real-world examples, evaluation criteria, and strengths and weaknesses.

3. Recognizing the immense potential of FOG computing at the network edge, a scholarly review [15] explores its versatile platform capabilities and emerging opportunities for services. The review advocates for FOG's ability to support latest programs and services, emphasizing its unique qualities. Viewing FOG as a foundational element for various applications, the review highlights its flexibility in enabling real-time decision-making for IoT, optimizing resource allocation in smart networks, facilitating autonomous operations in connected vehicles, enhancing traffic control in smart cities, and enabling edge data analysis in wireless sensor networks. To advance FOG computing further, the review emphasizes the importance of focusing on architecture development, orchestration and management of dispersed nodes, and the development of services and applications to fully utilize FOG's capabilities. This holistic perspective underscores FOG's potential as a transformative platform for innovative applications at the network edge.
4. According to a study [16], FOG computing has greatly impacted educational institutions by improving efficiency and creating better learning environments through the combination of IoT edge computing and wireless sensors. It helps monitor student interactions, classroom occupancy, and environmental conditions to optimize resource allocation. By utilizing CLOUD resources, FOG computing simplifies data storage and complex computations for collaborative assignments. It also plays a vital role in data analytics for educational datasets and promotes communication and resource sharing between teachers and students. However, challenges such as infrastructure investments and privacy concerns must be addressed. Emerging trends in education include advanced analytics, AR/VR integration, wearable technology, collaborative learning, and biometric sensors.

III. METHODOLOGY

A. System Model

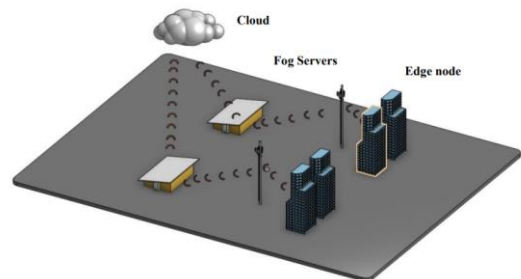


Fig. 1. Architecture of FOG Computing

Multiple sensors are installed throughout the house [9,10], to collect data such as temperature, light ambience levels, and total power consumed by the house. In the context of the power grid, collective data from all the houses is compiled for

the entire building. This data is meant to be processed, and all samples are correlated to power consumption. (**note- These solutions can be used to manage data more efficiently in the long run, and are not being commonly used yet)

The paper outlines methodologies that can be applied efficiently to accumulate and process data collected in building-integrated sensor-based environments, such as those in FOG computing. It explains how data collected and computed, enable real-time decision-making at the FOG layer regarding temperature, light ambience, and total power consumption.

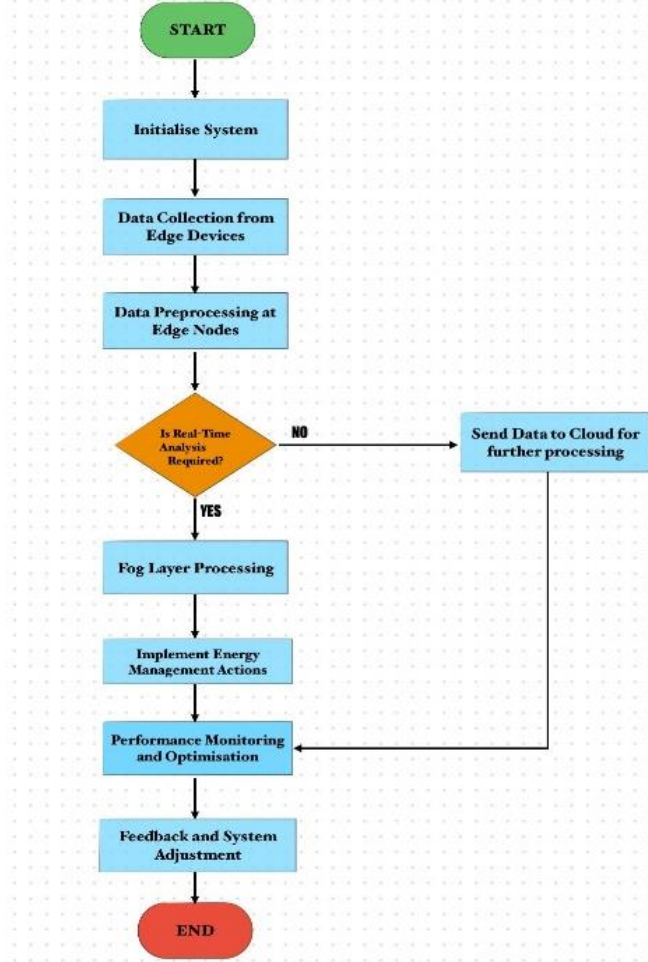


Fig.2. Representation of the architecture

B. Accumulation of data

1) Light Ambience

These sensors handle calculating required and used power to maintain a light in the on state. FOG nodes near the sensors process this data in real-time, capturing variations in light usage patterns.

2) Total Power Consumed

Instead of standard power meters, households can use smart power meters that monitor real-time power consumption data. This data is processed locally by FOG nodes at the edge, allowing for immediate insights into individual power usage and enhancing security by keeping critical information within the network. FOG nodes can also make decentralized decisions to refine power consumption without relying on a central grid management system. Each residential building is part of an edge layer and multiple

buildings in an area make up a FOG layer with a range of 200-250 km. The FOG layer has two functions [6]:

1. The primary purpose of the FOG layer is to process the data and send it to respective client swiftly. Since data at this level (Temperature, light ambience etc.) does not require high processing power FOG computing is used, resulting in reduced execution time.
2. The secondary function of the FOG layer is to decide the computational power required for high level data. The layer performs decision-making on whether data is to be sent to the CLOUD or can be processed at a FOG level [4]. The amount of power consumed for computing the data is also provided to the client.

3) iFogSim

iFogSim is a complex software framework built for FOG computing simulation and modelling. iFogSim, created to solve the issues of the Internet of Things (IoT) and edge computing, allows users to analyse and assess the performance of FOG computing architectures in a variety of scenarios. This comprehensive tool gives academics, developers, and decision-makers a platform to evaluate the impact of various parameters on system behaviour, assisting in the optimization of resource utilization in FOG computing environments. iFogSim helps in the understanding of FOG computing technologies and supporting innovation in the field of distributed and edge computing systems due to its flexible and adaptable design.

IV. ALGORITHM

Since there is a transmission of many bits, using the method of M-Ary Phase Shift Keying modulation is ideal. This method also determines whether the data transmitted will go to the FOG or the CLOUD.

$$2^N = M \quad (1)$$

$$\text{Phase Shift} = \frac{2\pi}{M} \quad (2)$$

The Duration of each bit T_s is given by:

$$T_s = NT_b \quad (3)$$

Transmitted waveform $S(t)$ given by:

$$S(t) = \sqrt{2P_s} \cos(2\pi fct + \phi_m) \quad (4)$$

$$\sqrt{2P_s} \cos(2\pi fct) \cos(\phi_m) - \sqrt{2P_s} \sin(2\pi fct) \sin(\phi_m) \quad (5)$$

Where ϕ_m is given by

$$\phi_m = \frac{(2m+1)\pi}{M} \quad (6)$$

$$s(t) = \sqrt{P_s T_s} \sqrt{\frac{2}{T_s}} \cos(2\pi fct) \cos(\phi_m) - \sqrt{P_s T_s} \sqrt{\frac{2}{T_s}} \sin(2\pi fct) \sin(\phi_m) \quad (7)$$

$$\phi_1(t) = \sqrt{\frac{2}{T_s}} \cos(2\pi fct) \quad (8)$$

$$\phi_2(t) = \sqrt{\frac{2}{T_s}} \sin(2\pi fct) \quad (9)$$

Therefore:

$$s(t) = \sqrt{P_s T_s} \cos(\phi_m) \phi_1(t) - \sqrt{P_s T_s} \sin(\phi_m) \phi_2(t) \quad (10)$$

Bandwidth will be given as:

$$BW = f_s - (-f_s) = 2f_s = \frac{2}{T_s} = \frac{2}{NT_b} = \frac{2T_b}{N} \quad (11)$$

If it does not be in this bandwidth then the data will go to FOG else, it will go to the CLOUD.

Energy of signal:

$$E_b = \int_0^{T_s} |S(t)|^2 dt \quad (12)$$

Where: $S(t)^2 = 2 E_b \cos^2(2\pi fct + \phi_m)$

$$E_b = \int_0^{T_s} 2 P_s \cos^2(2\pi fct + \phi_m) dt \quad (13)$$

We know that,

$$\cos^2(x) = \frac{1 + \cos 2x}{2} \quad (14)$$

So,

$$\cos^2(2\pi fct + \phi_m) = \frac{1 + \cos(4\pi fct + 2\phi_m)}{2} \quad (15)$$

$$E_b = \int_0^{T_s} 2 P_s \left\{ \frac{1 + \cos(4\pi fct + 2\phi_m)}{2} \right\} dt$$

$$E_b = P_s \int_0^{T_s} \{1 + \cos(4\pi fct + 2\phi_m)\} dt \quad (16)$$

Energy per Symbol:

$$E_b = E_b \{ T_s + 0 \} \quad (17)$$

$$E_b = \frac{P_s * T_s}{N} \quad (18)$$

Energy per Symbol:

$$E_s = E_b * N = 4 * E_b \quad (19)$$

For Power:

$$P_s = \frac{E_s}{T_s} = \frac{4 E_b}{T_s} \quad (20)$$

Time Delays:

Transmission Delay:

$$T_{trans} = \frac{L}{R} \quad (21)$$

Where “L” is Packet Length and “R” is Transmission Rate
Processing Delay:

$$T_{proc} = \frac{s}{p} \quad (22)$$

Where “s” is Size of data and “p” is Processing Speed
Queuing Delay:

$$T_{queue} = \frac{\lambda * T_{trans}}{1 - \rho} \quad (23)$$

Where λ is arrival rate of packets and ρ is Traffic intensity.

Propagation Delay:

$$T_{prop} = \frac{d}{v} \quad (24)$$

Where d is Distance between sender and receiver and v is propagation speed.

The time delay for edge is given as:

$$T_{edge} = T_{queue} + T_{prop} \quad (25)$$

The time delay for FOG is given as:

$$T_{fog} = T_{trans} + T_{proc} + T_{edge} \quad (26)$$

The time delay for CLOUD is given as:

$$T_{cloud} = T_{fog} + T_{back} \quad (27)$$

*Note: For FOG layer $T_{back} \approx 0$

V. SIMULATION

1) Pseudocode

(Note: - Expanding on one application of the smart home)

Input: Number of rooms (houses), devices per room (4 in this case)

Output: Execution time and network usage comparison between FOG and CLOUD deployment

1. Initialize

1. Create empty lists for Fog Devices, Sensors, Actuators.
2. Set the number of houses and devices per house.
3. Set transmission rate (5).
4. Choose deployment mode (Cloud = False, Fog = True).

2. Application Setup

1. Create an application with modules:
2. Connect modules with edges and specify data types:
3. Define selectivity for module outputs:
4. Define application loops to monitor latency:

3. Fog Network Setup

1. Create Fog Broker and Application.

2. Create Fog Devices for each room with specified characteristics (CPU, RAM, storage).
3. Create Sensors (Air Temperature sensor) and Actuators (Precision Temperature Control) for each room.
4. Define placement strategy for modules:
4. Simulation Run
 1. Initialize controller with Fog Devices, Sensors, Actuators.
 2. Submit application to the controller with chosen module placement strategy.
 3. Start CloudSim simulation.
 4. Collect and analyze data on latency and network usage for each loop in both Cloud and Fog modes.
5. Output
 5. Compare and analyze latency and network usage results for Fog and Cloud deployments.
 6. Draw conclusions about the efficiency and suitability of each approach for smart home systems with different device densities.

2) Simulation Results

TABLE I. EXECUTION TIME AND NETWORK USAGE TABULATION

Parameters		CLOUD		FOG	
No. of House per building	No. of devices per house	Executi on time	network usage	Executi on time	Network usage
3	4	845	484370	555	32512.8
3	6	831	725904.8	822	725904.8
3	8	1093	967439.6	1065	967439.6
3	10	1160	1015474.4	1158	1015474.4
4	4	826	645395	611	41650.4
4	6	1073	967441.4	1128	967441.4
4	8	1131	1018587.8	1188	1018587.8
4	10	1241	1031034.2	1263	1031034.2
3	4	979	806420	824	54868
3	6	1209	1015478	1172	1015478
3	8	1229	1031036	1191	1031036
3	10	1323	1046594	1390	1046594
6	4	1022	967445	868	66659.6
6	6	1224	1024814.6	1153	1024814.6
6	8	1248	1043484.2	1326	1043484.2
6	10	1313	1062153.8	1438	1062153.8
7	4	1103	1012370	1085	7429.2
7	6	1235	1034151.2	1255	1034151.2
7	8	1355	1055932.4	1463	1055932.4
7	10	1594	1077713.6	1606	1077713.6
8	4	1094	1018595	949	89626.8
8	6	1265	1043487.8	1334	1043487.8
8	8	1400	1068380.6	1419	1068380.6
8	10	1499	1093273.4	1535	1093273.4

The data presented above compares the efficiency and speed of FOG and CLOUD computing paradigms at the network's edge. To determine the performance characteristics of each architecture, a dataset that includes execution time, network use, and the number of devices per housing is used.

3) Execution Time

As the number of devices in each household increases, the time it takes for FOG and CLOUD computing to execute also increases. The execution time for FOG computing, for

example, goes up from 555 ms to 1158 ms, indicating a rise of 108.64%. On the other hand, CLOUD computing's execution time goes up from 845 ms to 1160 ms, an increase of 37.27%. This can be attributed to the growing number of devices in each household. Despite a significant increase in the number of connected devices (from 12 to 24), the execution time for FOG computing remains stable. This suggests that there are effective mechanisms in place to handle the increased workload efficiently without compromising on performance. Possible reasons for this could include:

- Scaling of the FOG infrastructure: Adding more FOG nodes to distribute the load.
- Caching of frequently used data: Reducing the need to process the same data repeatedly.
- Offloading processing tasks to connected devices: Sharing the processing load across multiple devices.
- Optimizing FOG nodes for specific applications: Tailoring FOG nodes to handle types of data or processes more efficiently.

FOG computing processes data near the devices, while CLOUD computing processes data in a centralized location. The longer distance data must travel to the CLOUD contributes to the higher execution time. However, the increase in execution time is much more noticeable for CLOUD computing compared to FOG computing.

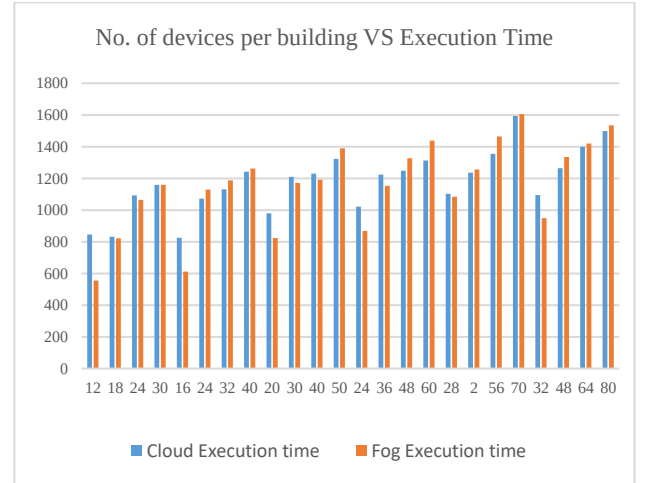


Fig. 3: Graphical representation of execution time

4) Network Usage

As the number of devices per household rises, so does the network usage for both FOG and CLOUD computing. This trend is natural, given that more devices are both generating and consuming data. The data shows a notable difference in the growth of network usage between FOG and CLOUD computing. While CLOUD usage grew at a modest pace, FOG usage saw a significant increase. This difference points to fundamental variations in the communication patterns and data handling strategies used by these two architectures. Reasons for this include:

- The more intensive data exchange between FOG nodes: The nodes might be communicating more frequently with devices, leading to a higher network usage.

- The CLOUD's use of data aggregation techniques: CLOUD computing might aggregate data before processing, reducing the volume of data transferred.

However, the increase in network usage is much more significant for FOG computing than for CLOUD computing. As the number of devices per household increases, FOG network usage surpasses that of CLOUD computing. This indicates that FOG computing can more efficiently use network resources for a smaller number of devices.

